# VF-TUF2000-S
Serisi | Series

**Debimetre**
Flowmeter

VFA ELEKTRONİK

# WALL MOUNTED ULTRASONIC FLOWMETER



clamp on

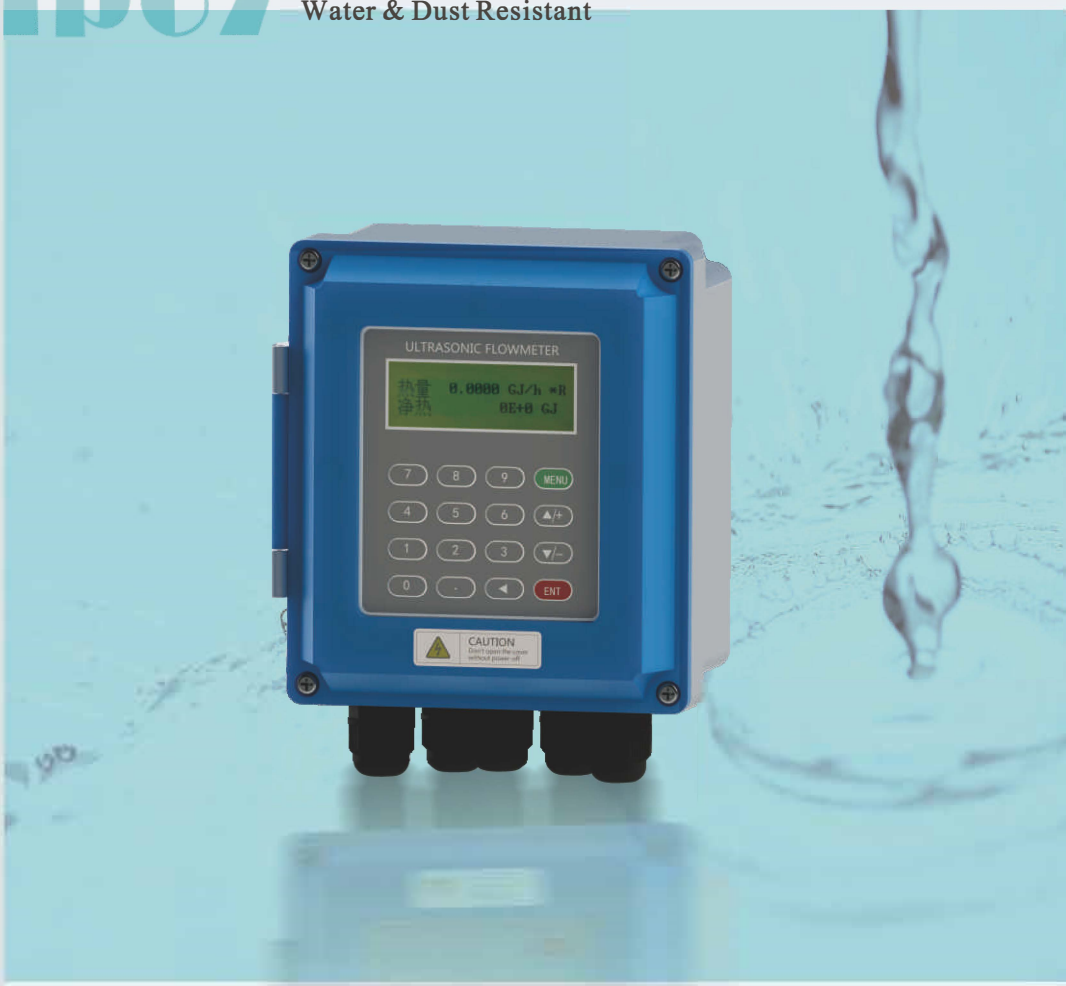


Insertion



Pipe

IP67 Water & Dust Resistant




## | Features |

- Measurement accuracy 1%.
- Convertor protection class IP67, transducer protection class IP68.
- Wide measurement range, pipe size from DN15mm to DN6000mm.
- Three types installation method: wall mounted , DIN-rail mounted , explosion-proof box mounted.
- Connect the temperature transducer can finish the heat/energy measurement.

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# TUF-2000B Wall Mounted Ultrasonic Flowmeter

VFA ELEKTRONİK

## | Introduction |

TUF-2000B flowmeter can be virtually applied to a wide range of in long-term online measurement. Convertor protection class IP67, transducer protection class IP68. A variety of liquid applications can be accommodated:ultra-pure liquids, portable water, chemicals, raw sewage, reclaimed water, cooling water, river water, plant effluent.etc

## | Measuring Diagram |

| Flow Measurement | Heat Measurement | Features |
|---|---|---|

### Calmp on






- No need to cut off water , no pressure loss.
- Connect clamp on temperature transducer,can finish the heat/energy measurement.
- Easy for installation.

### Insertion

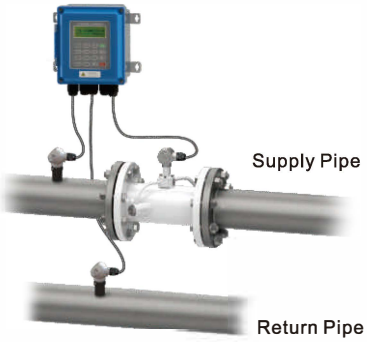





- No need to cut off water , no pressure loss.
- Stable and reliable for long term operation.
- Connect Pt100 temperature transducer, can finish the heat/energy measurement.

### Pipe






- Need to cut off pipe
- With high accuracy and stability
- Connect Pt100 temperature transducer, can finish the heat/energy measurement

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# TUF-2000B Wall Mounted Ultrasonic Flowmeter

## | Optional Transducers |

Please choose the suitable transducer,according to different liquid pipe condition and installation method, (please refer to the measuring diagram).

| Types | Picture | Spec. | Model | Measurement Range | Temperature | Dimension | |
|---|---|---|---|---|---|---|---|
| Clamp on |  | Small Size | TS-2 | DN25~DN100 | -30~90℃ | 45×25×32mm | |
| |  | Medium Size | TM-1 | DN50~DN700 | -30~90℃ | 64×39×44mm | |
| |  | Large Size | TL-1 | DN300~DN6000 | -30~90℃ | 97×54×53mm | |
| High temp clamp on |  | Small Size | TS-2-HT | DN25~DN100 | -30~160℃ | 45×25×32mm | |
| |  | Medium Size | TM-1-HT | DN50~DN700 | -30~160℃ | 64×39×44mm | |
| |  | Large Size | TL-1-HT | DN300~DN6000 | -30~160℃ | 97×54×53mm | |
| Insertion |  | Standard | TC-1 | DN80~DN6000 | -30~160℃ | 190×80×55mm | |
| |  | Lengthen | TC-2 | DN80~DN6000 | -30~160℃ | 335×80×55mm | |
| Pipe |  | π type | G1 | DN15~DN32 | -30~160℃ | SUS304 thread connection | Please refer to detailed pipe dimensions |
| |  | Standard | G2 | DN40~DN1000 | -30~160℃ | carbon steel thread connection | |

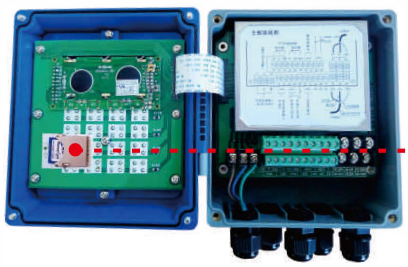
## | Optional Temperature Transducers |

Connect the temperature transducer can finish the heat/energy measurement

| Picture | Specification | Model | Meas. Range | Temperature | Cut of water | Accuracy |
|---|---|---|---|---|---|---|
|  | Clamp on temperature Transducer Pt100 | CT-1 | >DN50 | -40~160℃ | No | 100℃ ± 0.8℃ |
|  | Insertion temperature Transducer Pt100 | TCT-1 | >DN50 | -40~160℃ | Yes | |
|  | Insertion Pt100 Installation with pressure | PCT-1 | >DN50 | -40~160℃ | No | |
|  | Insertion Pt100 Small size pipe diameter | SCT-1 | <DN50 | -40~160℃ | Yes | |

## | Optional SD card memory |

Convenience for long-term and large data storage. Available to do the data processing like tabulation, statistical, data analysis, printing reports and production flow curve by using data analysis software.




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# TUF-2000B Wall Mounted Ultrasonic Flowmeter

## | Datasheet |

| Items | | Performance & Parameter |
|---|---|---|
| Convertor | Principle | Transit-time ultrasonic flowmeter |
| | Accuracy | ±1% |
| | Display | 2×20 character LCD with backlight, support the language of Chinese, English, Italy, Turkish |
| | Signal Output | 1 way 4~20mA output, electric resistance 0~1K，accuracy0. 1% |
| | | 1 way OCT pulse output（Pulse width ~1000ms，default is 200ms) |
| | | 1 way Relay output |
| | Signal Input | 3 way 4~20mA input, accuracy 0. 1%， acquisition signal such as temperature, press and liquid level |
| | | Connect the temperature transducer Pt100, can finish the heat/energy measurement |
| | Data Interface | Insulate Rs485 serial interface, upgrade the flowmeter software by computer, support the MODBUS |
| Special Cable | | Twisted-pair cable, generally, the length under 50 meters; Select the RS485, transmission distance can over 1000m |
| Pipe Installation Condition | Pipe Material | Steel, Stainless steel, Cast iron, Copper, Cement pipe, PVC, Aluminum, Glass steel product, liner is allowed |
| | Pipe Diameter | 15~6000mm |
| | Straight Pipe | Transducer installation should be satisfied: upstream10D, downstream 5D, 30D from the pump |
| Measuring Medium | Type of Liquid | Single liquid can transmit sound wave,<br>such as Water (hot water, chilled water, city water, sea water, waste water, etc.);<br>Sewage with small particle content; |
| | Temperature | -30~160℃ |
| | Turbidity | No more than 10000ppm and less bubble |
| | Flowrate | 0~±7m/s |
| Working Environment | Temperature | Convertor：-20~60℃；Flow Transducer：-30~160℃ |
| | Humidity | Convertor: 85% RH |
| Power Supply | | Dc8~36V or Ac85~264V（optional） |
| Power Consumption | | 1. 5W |
| Dimension | | 132*150*85 mm(convertor) |

## | Ordering Code |

TUF-2000B ----- [Flow Transducer] -DN [Diameter] mm - [Pipe Material] -- [Pressure] MPa - [Cable Length] m - [Temperature Transducer] ---------- [SD card Memory]

Flow Transducer:
TS-2
TM-1
TL-1
TS-2-HT
TM-1-HT
TL-1-HT
TC-1
TC-2
G

Pipe Material:
0 Carbon Steel
1 Stainless Steel
2 Cast Iron
3 Glass Fiber reinforced
4 PVC
5 Cement
6 Others

Temperature Transducer:
None
CT-1
TCT-1
PCT-1
SCT-1

SD card Memory:
0 Yes
1 No

Example TUF-2000B+TM-1+DN300+0+1. 6MPa+10m+PCT-1+0

PS: Standard confirguration=TUF-2000B(convertor)+TM-1+5M cable +Steel belt(DN200) +Couplant(1 pc)

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